

FIXED POINTS AND MULTISTABILITY IN MONOTONE BOOLEAN NETWORK MODELS

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MODELING GENE REGULATION

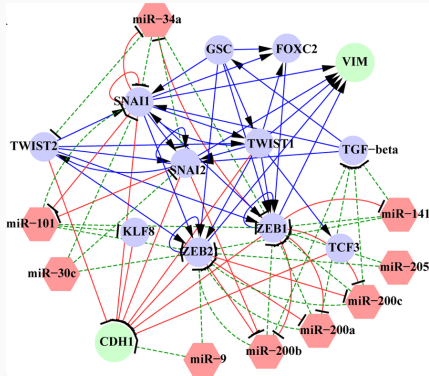


Figure: EMT network¹

¹Huang et al., *PLOS Computational Biology*, 2017

²Huang et al., *BMC Systems Biology*, 2018. Figure by Kishore Hari

³Albert and Thakar, *WIREs Systems Biology and Medicine*, 2014

MODELING GENE REGULATION

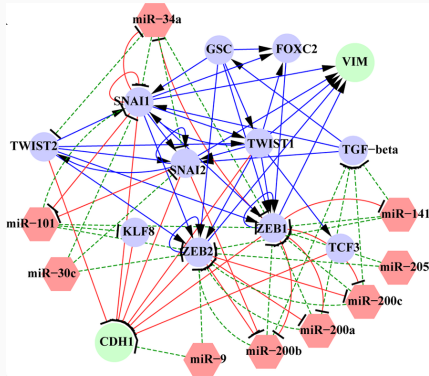


Figure: EMT network¹

ODE:

$$\frac{dX_1}{dt} = g_1 H^{s^+}(\cdot) H^{s^-}(\cdot) - k_1 X_1$$

$\vdots$

$$\frac{dX_N}{dt} = g_N H^{s^+}(\cdot) H^{s^-}(\cdot) - k_N X_N$$

¹Huang et al., *PLOS Computational Biology*, 2017

²Huang et al., *BMC Systems Biology*, 2018. Figure by Kishore Hari

³Albert and Thakar, *WIREs Systems Biology and Medicine*, 2014

MODELING GENE REGULATION

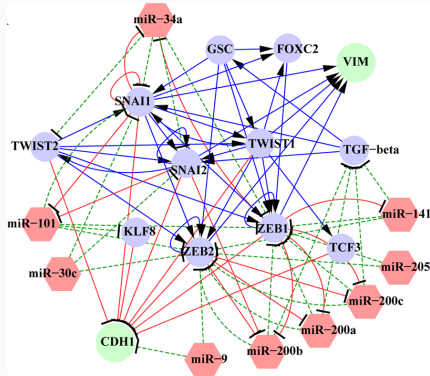


Figure: EMT network¹

¹Huang et al., *PLOS Computational Biology*, 2017

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³Albert and Thakar, *WIREs Systems Biology and Medicine*, 2014

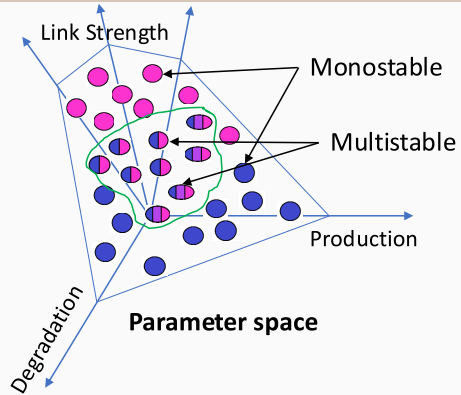


Figure: RACIPE: Parameter sampling²

MODELING GENE REGULATION

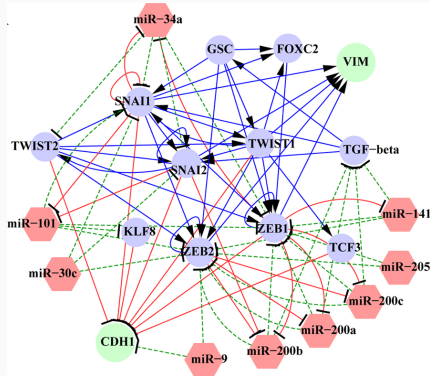


Figure: EMT network¹

Boolean³:

$$X_1(t+1) = X_1(t) \vee X_2(t) \cdots \wedge X_n(t)$$

$\vdots$

$$X_N(t+1) = \neg X_2(t) \cdots \vee \neg X_{n-1}(t)$$

¹Huang et al., *PLOS Computational Biology*, 2017

²Huang et al., *BMC Systems Biology*, 2018. Figure by Kishore Hari

³Albert and Thakar, *WIREs Systems Biology and Medicine*, 2014

Definition

A Boolean function $f: \mathbb{B}^k \rightarrow \mathbb{B}$ is *increasing with respect to input j* if for any input $b = (b_1, \dots, b_k) \in \mathbb{B}^k$

$$f(b_1, \dots, b_{j-1}, 0, b_{j+1}, \dots, b_k) \leq f(b_1, \dots, b_{j-1}, 1, b_{j+1}, \dots, b_k).$$

Definition

A Boolean function $f: \mathbb{B}^k \rightarrow \mathbb{B}$ is *monotone* if it is increasing or decreasing with respect to each input $i = 1, \dots, k$. In this case, f is called a *monotone Boolean function*.

A set of increasing monotone Boolean functions $f: \mathbb{B}^k \rightarrow \mathbb{B}$ will be denoted $MBF(k)$.

Definition

A Boolean model $f: \mathbb{B}^N \rightarrow \mathbb{B}^N, f = (f_1, \dots, f_N)$, is *monotone* if for every i the function f_i is monotone.

¹Gedeon, *npj Systems Biology and Applications*, 2024; Korshunov, *Russian Mathematical Surveys*, 2003

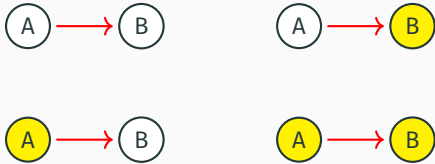


Figure: Schematic of increasing MBF(1) for activatory input

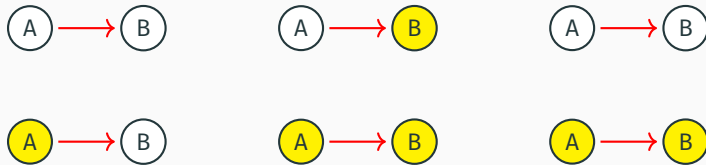


Figure: Schematic of increasing MBF(1) for activatory input

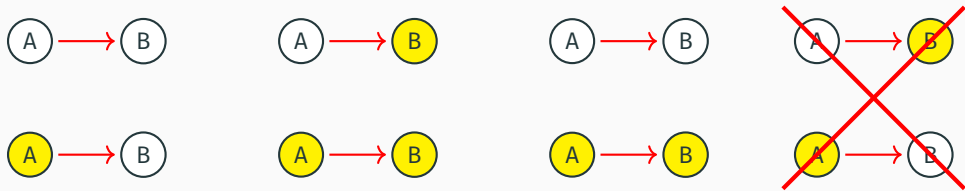


Figure: Schematic of increasing MBF(1) for activatory input

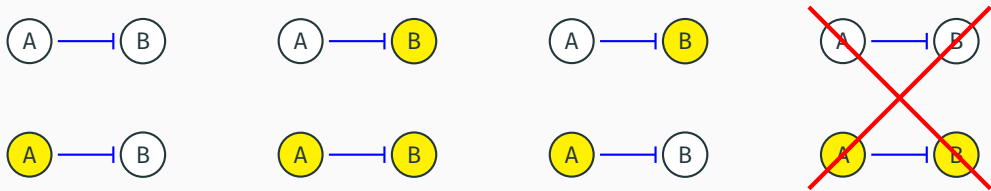


Figure: Schematic of decreasing MBF(1) for inhibitory input

Inputs		BFs															
X	Y																
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1

Table: Table of BF's for a node with two inputs X and Y.

Inputs		BFs															
X	Y																
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1

Table: Table of BF's for a node with two inputs X and Y.

¹Red indicates non-increasing functions (non-monotonous)

Inputs		MBFs					
X	Y	0	$\vee$	X	Y	$\wedge$	1
0	0	0	0	0	0	0	1
0	1	0	0	0	1	1	1
1	0	0	0	1	0	1	1
1	1	0	1	1	1	1	1

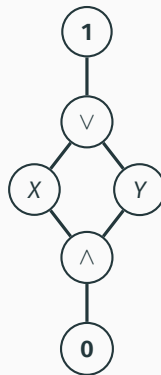


Figure: Table (left) and Lattice (right) of MBF(2) for a node with two activatory inputs X and Y.

$$\begin{array}{ccc} \mathbb{B}^N & \xrightarrow{f} & \mathbb{B}^N \\ \alpha \uparrow \downarrow \alpha & & \alpha \uparrow \downarrow \alpha \\ \mathbb{B}^N & \xrightarrow{g} & \mathbb{B}^N \end{array}$$

- $f = (f_1, \dots, f_N)$: NL-free network MBM
- α : Change of variable transformation
- $g = (g_1, \dots, g_N)$: Positive network MBM

$$\begin{array}{ccc} \mathbb{B}^N & \xrightarrow{f} & \mathbb{B}^N \\ \alpha \uparrow \downarrow \alpha & & \alpha \uparrow \downarrow \alpha \\ \mathbb{B}^N & \xrightarrow{g} & \mathbb{B}^N \end{array}$$

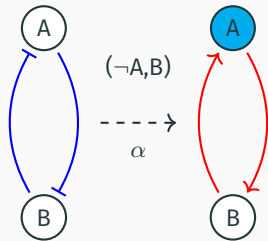
Trajectories and Steady states preserved

$$x = f(x) \iff \alpha(x) = g(\alpha(x)).$$

- $f = (f_1, \dots, f_N)$: NL-free network MBM
- α : Change of variable transformation
- $g = (g_1, \dots, g_N)$: Positive network MBM

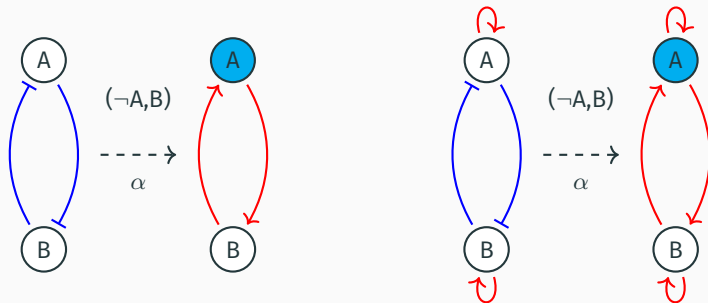
CHANGE OF VARIABLE - TOGGLE SWITCH





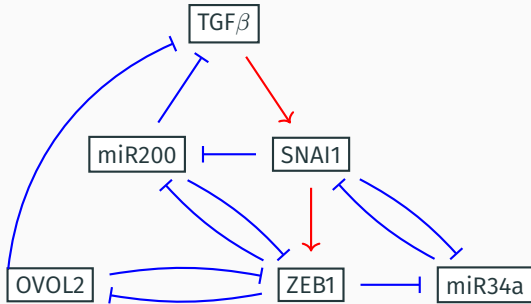
¹Gardner et al., *Nature*, 2000

CHANGE OF VARIABLE - TOGGLE SWITCH

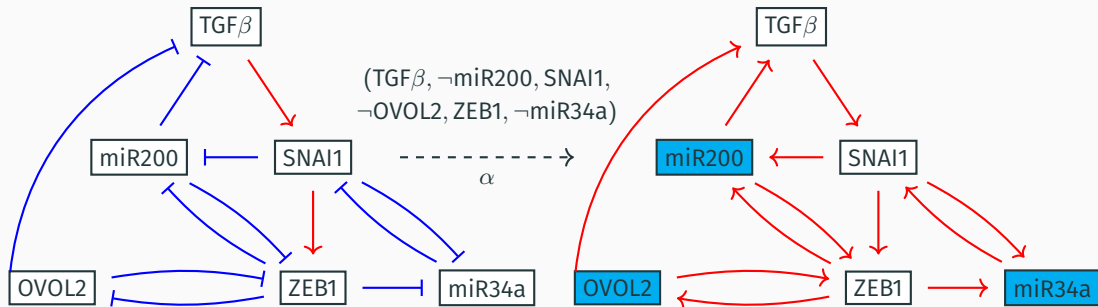


¹Gardner et al., *Nature*, 2000

CHANGE OF VARIABLE - EMT NETWORK



CHANGE OF VARIABLE - EMT NETWORK



¹Hong et al., *PLOS Computational Biology*, 2015

$$U(b) := \{f \in MBF(k) \mid f(b) = 1\}$$

$$L(b) := \{f \in MBF(k) \mid f(b) = 0\}.$$

¹ $\mathbf{o}_k$: vector of zeros of length k , $\mathbf{1}_k$: vector of ones of length k

$$U(b) := \{f \in MBF(k) \mid f(b) = 1\}$$

$$L(b) := \{f \in MBF(k) \mid f(b) = 0\}.$$

L(11)	$\{\mathbf{0}\}$	1
L(10)	$\{\mathbf{0}, \wedge, Y\}$	3
L(01)	$\{\mathbf{0}, \wedge, X\}$	3
L(00)	$\{\mathbf{0}, \wedge, X, Y, \vee\}$	5
U(11)	$\{\mathbf{1}, \vee, X, Y, \wedge\}$	5
U(10)	$\{\mathbf{1}, \vee, X\}$	3
U(01)	$\{\mathbf{1}, \vee, Y\}$	3
U(00)	$\{\mathbf{1}\}$	1

Figure: U and L for $MBF(2)$

¹ $\mathbf{o}_k$: vector of zeros of length k , $\mathbf{1}_k$: vector of ones of length k

$$U(b) := \{f \in MBF(k) \mid f(b) = 1\}$$

$$L(b) := \{f \in MBF(k) \mid f(b) = 0\}.$$

Lattice theory:

U s = prime ideals,

L s = prime filters

$L(11)$	$\{0\}$	1
$L(10)$	$\{0, \wedge, Y\}$	3
$L(01)$	$\{0, \wedge, X\}$	3
$L(00)$	$\{0, \wedge, X, Y, \vee\}$	5
$U(11)$	$\{1, \vee, X, Y, \wedge\}$	5
$U(10)$	$\{1, \vee, X\}$	3
$U(01)$	$\{1, \vee, Y\}$	3
$U(00)$	$\{1\}$	1

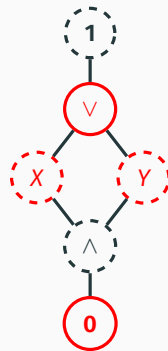


Figure: U and L for $MBF(2)$

¹ 0_k : vector of zeros of length k , 1_k : vector of ones of length k

$$U(b) := \{f \in MBF(k) \mid f(b) = 1\}$$

$$L(b) := \{f \in MBF(k) \mid f(b) = 0\}.$$

Lattice theory:

U s = prime ideals,

L s = prime filters

$L(\mathbf{o}_k)$ and $U(\mathbf{1}_k)$ are the largest sets
with size $|MBF(k)| - 1$

$L(11)$	$\{\mathbf{0}\}$	1
$L(10)$	$\{\mathbf{0}, \wedge, Y\}$	3
$L(01)$	$\{\mathbf{0}, \wedge, X\}$	3
$L(00)$	$\{\mathbf{0}, \wedge, X, Y, \vee\}$	5
$U(11)$	$\{\mathbf{1}, \vee, X, Y, \wedge\}$	5
$U(10)$	$\{\mathbf{1}, \vee, X\}$	3
$U(01)$	$\{\mathbf{1}, \vee, Y\}$	3
$U(00)$	$\{\mathbf{1}\}$	1

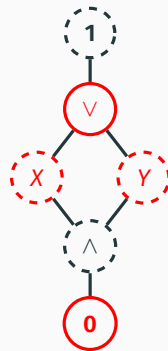


Figure: U and L for $MBF(2)$

¹ $\mathbf{o}_k$: vector of zeros of length k , $\mathbf{1}_k$: vector of ones of length k

MBMs for Positive network

$$MBM := MBF(k_1) \times MBF(k_2) \times \dots MBF(k_N)$$

¹ $\mathbf{S}(v_j)$ = input nodes of v_j , $k_j = |\mathbf{S}(v_j)|$, $e_{\mathbf{S}(j)}$ = values of $\mathbf{S}(v_j)$

MBMs for Positive network

$$MBM := MBF(k_1) \times MBF(k_2) \times \dots MBF(k_N)$$

Boolean state $e = (e_1, \dots, e_N) \in \mathbb{B}^N$

$$f_i \in Z_i(e)$$
$$Z_i(e) = \begin{cases} U(e_{\mathbf{S}(i)}) & \text{if } e_i = 1 \\ L(e_{\mathbf{S}(i)}) & \text{if } e_i = 0 \end{cases}$$

¹ $\mathbf{S}(v_j)$ = input nodes of v_j , $k_j = |\mathbf{S}(v_j)|$, $e_{\mathbf{S}(i)}$ = values of $\mathbf{S}(v_j)$

MBMs for Positive network

$$MBM := MBF(k_1) \times MBF(k_2) \times \dots MBF(k_N)$$

Boolean state $e = (e_1, \dots, e_N) \in \mathbb{B}^N$

$$f_i \in Z_i(e)$$

$$Z_i(e) = \begin{cases} U(e_{\mathbf{S}(i)}) & \text{if } e_i = 1 \\ L(e_{\mathbf{S}(i)}) & \text{if } e_i = 0 \end{cases}$$

MBMs that supports $\mathbf{o}$

$$L(\mathbf{o}_{k_1}) \times L(\mathbf{o}_{k_2}) \times \dots L(\mathbf{o}_{k_N}),$$

MBMs that supports $\mathbf{1}$

$$U(\mathbf{1}_{k_1}) \times U(\mathbf{1}_{k_2}) \times \dots U(\mathbf{1}_{k_N}).$$

Size

$$(|MBF(k_1)| - 1) \times (|MBF(k_2)| - 1) \times \dots (|MBF(k_N)| - 1). \quad (1)$$

¹ $\mathbf{S}(v_j)$ = input nodes of v_j , $k_j = |\mathbf{S}(v_j)|$, $e_{\mathbf{S}(i)}$ = values of $\mathbf{S}(v_j)$

STEADY STATES - TOGGLE SWITCH

$$U(1) = \{1, Id\}, U(0) = \{1\}, L(0) = \{0, Id\}, L(1) = \{0\}.$$

Positive Toggle switch:

- Total: $f \in MBF(1) \times MBF(1) = 9$ models

1	01	11	11
Id	00	11 00	11
0	00	00	10
	0	Id	1

Figure: Steady states supported by $MBF(1) \times MBF(1)$

¹Models supporting steady states: - - - (00), - · - (11)

$$U(1) = \{1, Id\}, U(0) = \{1\}, L(0) = \{0, Id\}, L(1) = \{0\}.$$

Positive Toggle switch:

- Total: $f \in MBF(1) \times MBF(1) = 9$ models
- (00):

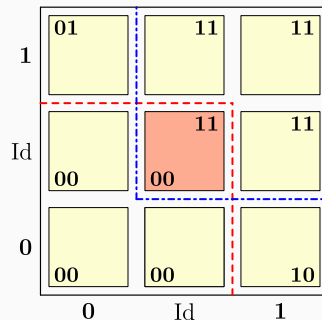


Figure: Steady states supported by $MBF(1) \times MBF(1)$

¹Models supporting steady states: - - - (00), - · - (11)

$$U(1) = \{1, Id\}, U(0) = \{1\}, L(0) = \{0, Id\}, L(1) = \{0\}.$$

Positive Toggle switch:

- Total: $f \in MBF(1) \times MBF(1) = 9$ models
- (00) : $f \in L(0) \times L(0) =$

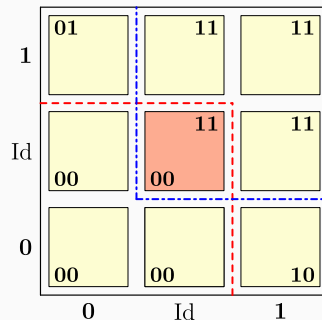


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STEADY STATES - TOGGLE SWITCH

$$U(1) = \{1, Id\}, U(0) = \{1\}, L(0) = \{0, Id\}, L(1) = \{0\}.$$

Positive Toggle switch:

- Total: $f \in MBF(1) \times MBF(1) = 9$ models
- (00): $f \in L(0) \times L(0) = 4$ models

1	01	11	11
Id	00	11 00	11
0	00	00	10
	0	Id	1

Figure: Steady states supported by $MBF(1) \times MBF(1)$

¹Models supporting steady states: - - - (00), - · - (11)

$$U(1) = \{1, Id\}, U(0) = \{1\}, L(0) = \{0, Id\}, L(1) = \{0\}.$$

Positive Toggle switch:

- Total: $f \in MBF(1) \times MBF(1) = 9$ models
- (00): $f \in L(0) \times L(0) = 4$ models
- (01): $f \in L(1) \times U(0) = 1$ model

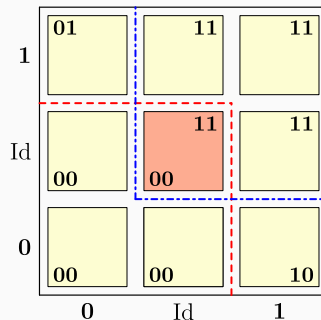


Figure: Steady states supported by $MBF(1) \times MBF(1)$

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$$U(1) = \{1, Id\}, U(0) = \{1\}, L(0) = \{0, Id\}, L(1) = \{0\}.$$

Positive Toggle switch:

- Total: $f \in MBF(1) \times MBF(1) = 9$ models
- (00): $f \in L(0) \times L(0) = 4$ models
- (01): $f \in L(1) \times U(0) = 1$ model
- (10): $f \in U(0) \times L(1) = 1$ model
- (11): $f \in U(1) \times U(1) = 4$ models

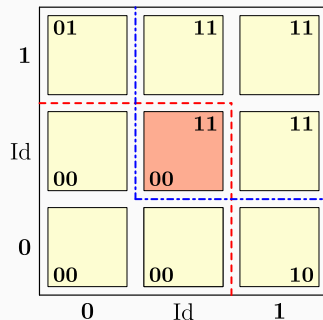


Figure: Steady states supported by $MBF(1) \times MBF(1)$

¹Models supporting steady states: - - - (00), - · - (11)

STEADY STATES - TOGGLE SWITCH

Toggle switch:

- Total: 9 models
- (10): 4 models
- (11): 1 model
- (00): 1 model
- (01): 4 models

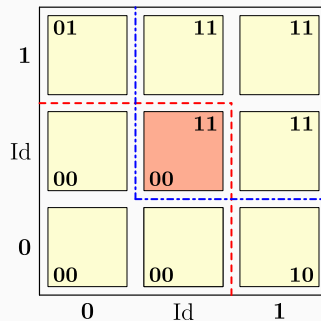


Figure: Steady states supported by $MBF(1) \times MBF(1)$

¹Models supporting steady states: - - - (00), - · - (11)

STEADY STATES - TOGGLE SWITCH WITH SELF-ACTIVATION

Size	(11)	(10)	(01)	(00)
L	1	3	3	5
U	5	3	3	1

Positive Toggle switch with self-activation:

- Total: $f \in MBF(2) \times MBF(2) = 36$ models

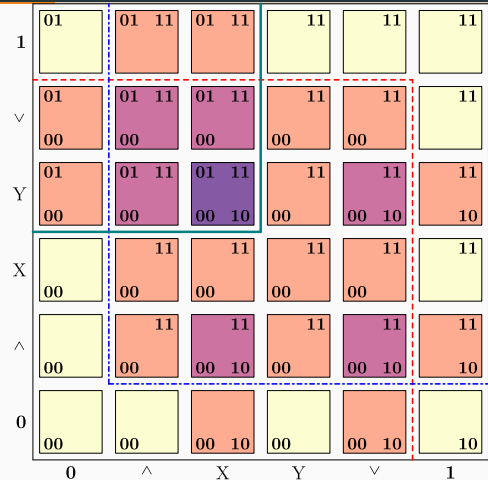


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Models supporting steady states: - - - (00), — (01), - · - (11)

STEADY STATES - TOGGLE SWITCH WITH SELF-ACTIVATION

Size	(11)	(10)	(01)	(00)
L	1	3	3	5
U	5	3	3	1

Positive Toggle switch with self-activation:

- Total: $f \in MBF(2) \times MBF(2) = 36$ models
- (00):

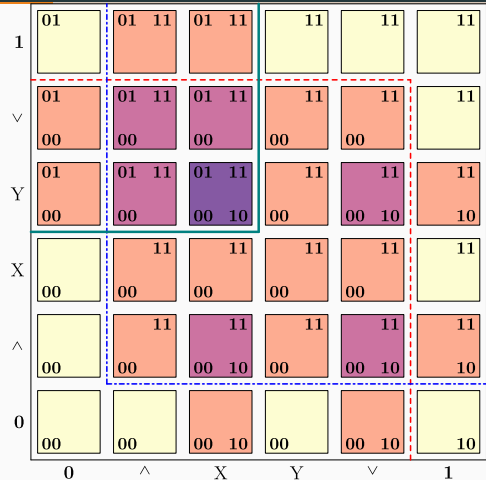


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Models supporting steady states: - - - (00), — (01), - · - · (11)

STEADY STATES - TOGGLE SWITCH WITH SELF-ACTIVATION

Size	(11)	(10)	(01)	(00)
L	1	3	3	5
U	5	3	3	1

Positive Toggle switch with self-activation:

- Total: $f \in MBF(2) \times MBF(2) = 36$ models
- (00): $f \in L(00) \times L(00) =$

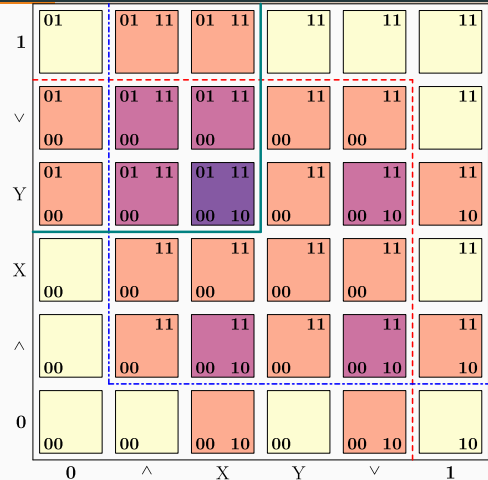


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Models supporting steady states: - - - (00), — (01), - · - · (11)

STEADY STATES - TOGGLE SWITCH WITH SELF-ACTIVATION

Size	(11)	(10)	(01)	(00)
L	1	3	3	5
U	5	3	3	1

Positive Toggle switch with self-activation:

- Total: $f \in MBF(2) \times MBF(2) = 36$ models
- (00): $f \in L(00) \times L(00) = 25$ models

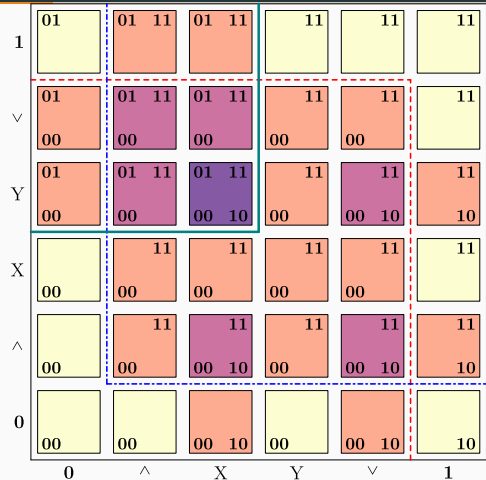


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Models supporting steady states: - - - (00), — (01), - · - (11)

STEADY STATES - TOGGLE SWITCH WITH SELF-ACTIVATION

Size	(11)	(10)	(01)	(00)
L	1	3	3	5
U	5	3	3	1

Positive Toggle switch with self-activation:

- Total: $f \in MBF(2) \times MBF(2) = 36$ models
- (00): $f \in L(00) \times L(00) = 25$ models
- (01): $f \in L(01) \times U(01) = 9$ model

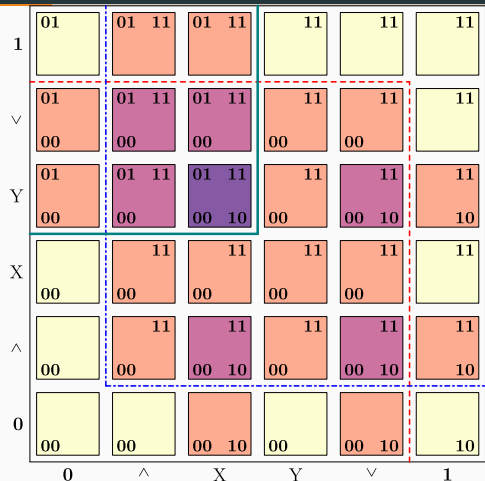


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Models supporting steady states: - - - (00), — (01), - . - (11)

STEADY STATES - TOGGLE SWITCH WITH SELF-ACTIVATION

Size	(11)	(10)	(01)	(00)
L	1	3	3	5
U	5	3	3	1

Positive Toggle switch with self-activation:

- Total: $f \in MBF(2) \times MBF(2) = 36$ models
- (00): $f \in L(00) \times L(00) = 25$ models
- (01): $f \in L(01) \times U(01) = 9$ model
- (10): $f \in U(10) \times L(11) = 9$ model
- (11): $f \in U(11) \times U(11) = 25$ models

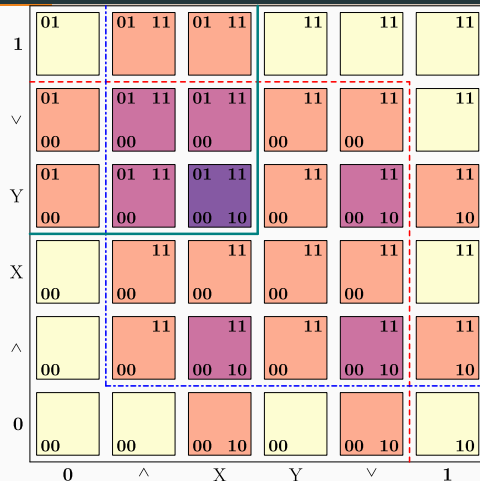


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Models supporting steady states: - - - (00), — (01), - · - (11)

STEADY STATES - TOGGLE SWITCH WITH SELF-ACTIVATION

Toggle switch with self activation:

- Total: 36 models
- (10): 25 models
- (00): 9 model
- (11): 9 model
- (01): 25 models

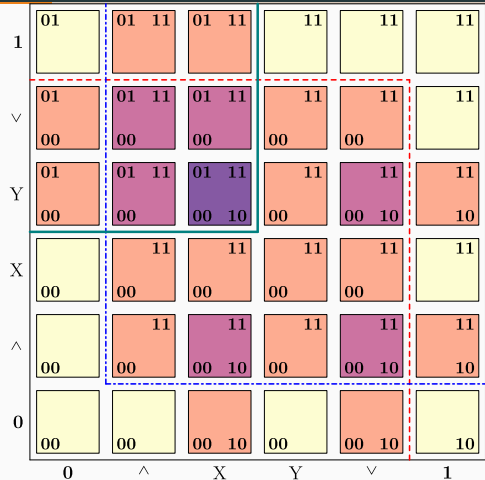


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Models supporting steady states: - - - (00), — (01), - · - (11)

State	Parameter sets	MBMs
Total	$MBF(2) \times MBF(2) \times MBF(2) \times MBF(1) \times MBF(3) \times MBF(2)$	77760
(000000)	$L(00) \times L(00) \times L(00) \times L(0) \times L(000) \times L(00)$	23750
(111111)	$U(11) \times U(11) \times U(11) \times U(1) \times U(111) \times U(11)$	23750
(000100)	$L(01) \times L(00) \times L(00) \times U(0) \times L(001) \times L(00)$	5250
(111011)	$U(10) \times U(11) \times U(11) \times L(1) \times U(110) \times U(11)$	5250
(001001)	$L(00) \times L(10) \times U(01) \times L(0) \times L(010) \times U(10)$	3780
(110110)	$U(11) \times U(01) \times L(10) \times U(1) \times U(101) \times L(01)$	3780
(100100)	$U(01) \times L(00) \times L(10) \times U(0) \times L(001) \times L(00)$	3150
(011011)	$L(10) \times U(11) \times U(01) \times L(1) \times U(110) \times U(11)$	3150

Table: Number of MBMs supporting the top 8 most common steady states.

¹E: Epithelial, M: Mesenchymal, I: Intermediate

State	MBMs	EMT-state	
Total	77760		
(000000)	23750	miR200 ⁺ OVOL2 ⁺ miR34a ⁺	E
(111111)	23750	TGF β ⁺ SNAI1 ⁺ ZEB1 ⁺	M
(000100)	5250	miR200 ⁺ miR34a ⁺	I
(111011)	5250	TGF β ⁺ SNAI1 ⁺ OVOL2 ⁺ ZEB1 ⁺	I
(001001)	3780	miR200 ⁺ SNAI1 ⁺ OVOL2 ⁺	I
(110110)	3780	TGF β ⁺ ZEB1 ⁺ miR34a ⁺	I
(100100)	3150	TGF β ⁺ miR200 ⁺ miR34a ⁺	I
(011011)	3150	SNAI1 ⁺ OVOL2 ⁺ ZEB1 ⁺	I

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Steady states: $d, e \in \mathbb{B}^N$

¹ $d_{S(i)}$ = values of the input nodes of v_i evaluated at the state d

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$$f_i \in Z_i(d) \cap Z'_i(e)$$
$$Z_i(d) = \begin{cases} U(d_{\mathbf{s}(i)}) & \text{if } d_i = 1 \\ L(d_{\mathbf{s}(i)}) & \text{if } d_i = 0 \end{cases}$$

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$$d_i = e_i \implies Z_i(d) \cap Z'_i(e) \neq \emptyset.$$

$$d_i \neq e_i \implies Z_i(d) \cap Z'_i(e) \text{ may be empty}$$

Note: Bistability between d, e : d and e among the steady states; others may also exist.

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Note: Bistability between d, e : d and e among the steady states; others may also exist.

Most common 0 and 1

$$\prod_{i=1}^N |MBF(k_i)| - 2$$

¹ $d_{\mathbf{s}(i)}$ = values of the input nodes of v_i evaluated at the state d

Steady states: $d^1, \dots, d^q \in \mathbb{B}^N$

$$f_i \in \bigcap_{j=1}^q Z_i(d^j)$$

$$Z_i(d^j) = \begin{cases} U(d_{\mathbf{s}(i)}^j) & \text{if } d_i^j = 1 \\ L(d_{\mathbf{s}(i)}^j) & \text{if } d_i^j = 0 \end{cases},$$

¹ $d_{\mathbf{s}(i)}^j$ = values of the input nodes of v_i evaluated at the state d^j

² $m_n(E)$ = MBMs supporting n –multistability for $E := \{e_1, \dots, e_n\}$

³ $em_q(A)$ = MBMs supporting exact q –multistability for $A = \{e_1, \dots, e_q\}$

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Exact q -multistability: supports exactly q -steady states

$$em_q(A) = m_q(A) \setminus \bigcup_{n>q, A \subseteq E} m_n(E)$$

¹ $d_{\mathbf{s}(i)}^j$ = values of the input nodes of v_i evaluated at the state d^j

² $m_n(E)$ = MBMs supporting n -multistability for $E := \{e_1, \dots, e_n\}$

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Positive Toggle switch:

$$d = (00) \quad \text{and} \quad e = (11)$$

$$f_1 \in L(0) \cap U(1) \quad \text{and} \quad f_1 \in L(0) \cap U(1) \\ (Id, Id)$$

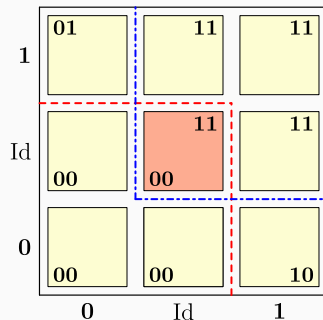


Figure: Steady states supported by $MBF(1) \times MBF(1)$

¹ Monostable, Bistable

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$$d = (00) \quad \text{and} \quad e = (11)$$

$$f_1 \in L(0) \cap U(1) \quad \text{and} \quad f_1 \in L(0) \cap U(1) \\ (Id, Id)$$

$$d = (00) \quad \text{and} \quad e = (10)$$

$$f_1 \in L(0) \cap U(0) \quad \text{and} \quad f_2 \in L(0) \cap L(1)$$

$$L(0) \cap U(0) = \emptyset \rightarrow \text{no MBMs}$$

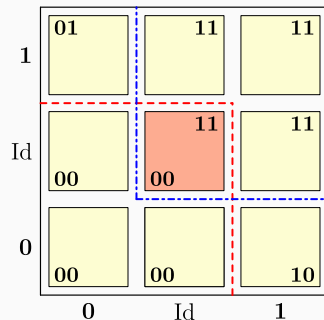


Figure: Steady states supported by $MBF(1) \times MBF(1)$

¹ Monostable, Bistable

BISTABILITY - TOGGLE SWITCH

Toggle switch:

- (10) and (01): 1 model

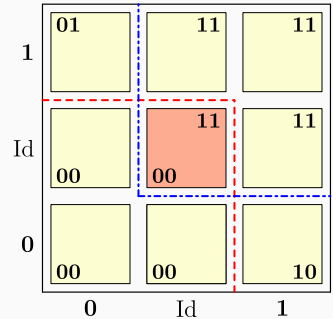


Figure: Steady states supported by $MBF(1) \times MBF(1)$

¹ Monostable, Bistable

Bistability between E and M states

$$4 \times 4 \times 4 \times 1 \times 18 \times 4 = 4608 \text{ MBMs}$$

$$\text{Prevalence} = 4608/77760 = 0.059 \approx 6\%$$

BISTABILITY & MULTISTABILITY - TOGGLE SWITCH WITH SELF-ACTIVATION

Positive Toggle switch with self-activation:

$$d = (00) \quad \text{and} \quad e = (11)$$

$$(L(00) \cap U(11)) \times (L(00) \cap U(11)) = 16 \text{ models}$$

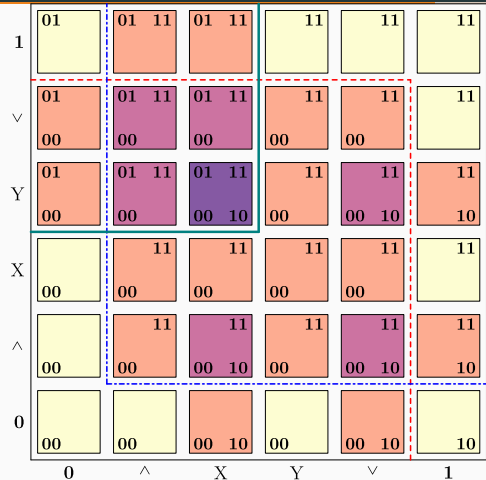


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Exact multistability: ■ Monostable, ■ Bistable, ■ Tristable, ■ Quadrastable

BISTABILITY & MULTISTABILITY - TOGGLE SWITCH WITH SELF-ACTIVATION

Positive Toggle switch with self-activation:

$$d = (00) \quad \text{and} \quad e = (11)$$

$$(L(00) \cap U(11)) \times (L(00) \cap U(11)) = 16 \text{ models}$$

- (00) and (01): 6 models
- (00) and (10): 6 models
- (01) and (11): 6 models
- (10) and (11): 6 models
- (01) and (10): 1 model

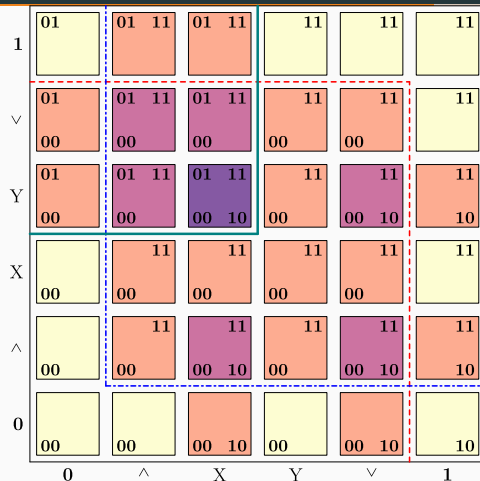


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Exact multistability: ■ Monostable, ■ Bistable, ■ Tristable, ■ Quadrastable

BISTABILITY & MULTISTABILITY - TOGGLE SWITCH WITH SELF-ACTIVATION

Toggle switch with self-activation:

- (10) and (01): 16 models
- (10) and (11): 6 models
- (10) and (00): 6 models
- (11) and (01): 6 models
- (00) and (01): 6 models
- (11) and (00): 1 model

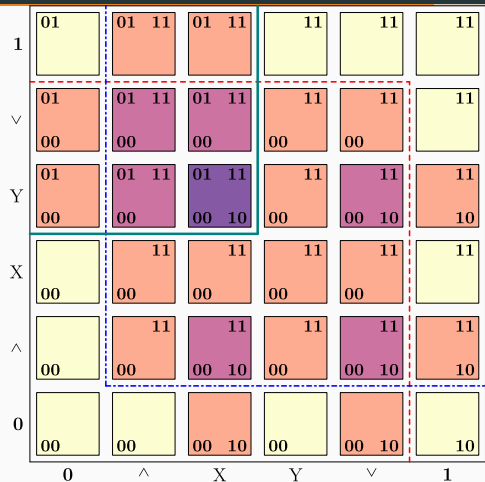


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Exact multistability: ■ Monostable, ■ Bistable, ■ Tristable, ■ Quadrastable

BISTABILITY & MULTISTABILITY - TOGGLE SWITCH WITH SELF-ACTIVATION

Positive Toggle switch with self-activation:

- Tristability:
 - (00, 01, 11): 4 models
 - (00, 10, 11): 4 models
- Quadrastability:
 - (00, 10, 01, 11): 1 model

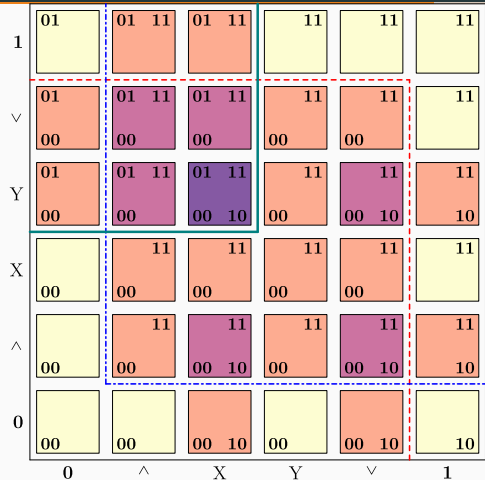


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Exact multistability: Monostable, Bistable, Tristable, Quadrastable

BISTABILITY & MULTISTABILITY - TOGGLE SWITCH WITH SELF-ACTIVATION

Toggle switch with self-activation:

- Tristability:
 - (10, 11, 01): 4 models
 - (10, 00, 01): 4 models
- Quadrastability:
 - (10, 00, 11, 01): 1 model

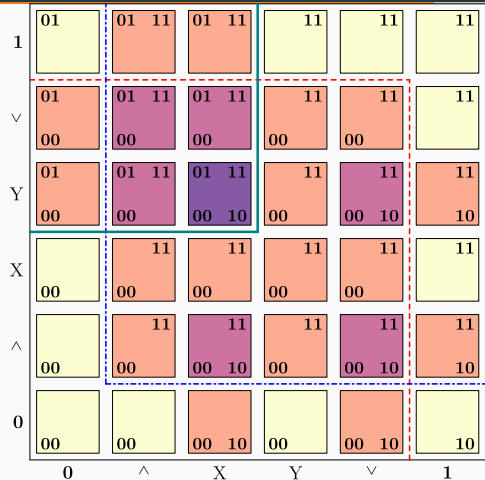


Figure: Steady states supported by $MBF(2) \times MBF(2)$

¹Exact multistability: ■ Monostable, ■ Bistable, ■ Tristable, ■ Quadrastable

- Mathematical framework: explore GRN dynamics without simulations

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- Mathematical framework: explore GRN dynamics without simulations
- Team networks (NL-free) most common states:
 - A high, B low
 - A low, B high
 - Regardless of network size, density as long as coherent
- Bistability highest between these states

- Negative loop influence - Complex attractors
- Designing networks with certain properties

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THANK YOU



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